ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 4 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy practice exam 4.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept any correct, equivalent code or pseudocode. Do not penalise small syntax slips (e.g. a missing semicolon) more than once per question unless syntax is being tested. Give follow-through marks.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 B ✓ 1.1.9 A ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 E ✓ 1.3.4 B ✓ 1.3.5 I ✓ 1.3.6 D ✓ 1.3.7 H ✓ 1.3.8 A ✓ 1.3.9 J ✓ 1.3.10 F ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Input voltage ✓; IF voltage < 6,5 ✓ display "Charge now" ✓; ELSE IF voltage < 7,0 ✓ display "Low" ✓; ELSE display "OK" / end ✓ (6)

2.2 Declare a float ✓; prompt ✓ and scanf("%f", &v) ✓; first if ✓; else if ✓; else ✓; correct messages ✓; main / return / braces ✓, e.g. (8)

```
#include <stdio.h>

int main() {
  float v;
  printf("Battery voltage: ");
  scanf("%f", &v);
  if (v < 6.5) {
    printf("Charge now");
  } else if (v < 7.0) {
    printf("Low");
  } else {
    printf("OK");
  }
  return 0;
}
```

2.3 Any FOUR sensible values with correct outputs ✓✓✓✓, including boundaries, e.g. 6,0 → Charge now; 6,5 → Low; 6,9 → Low; 7,0 → OK; 8,4 → OK (4)

2.4 Mistakes often happen exactly at the limits (e.g. using < instead of <=) ✓, so testing the boundary values shows whether the program decides correctly there ✓. (2)

[20]

QUESTION 3

3.1 float array of 5 ✓; for loop 0 to 4 ✓; prompt ✓; scanf with &laps[i] ✓; correct syntax ✓, e.g. (5)

```
float laps[5];
for (int i = 0; i < 5; i++) {
  printf("Lap %d time: ", i + 1);
  scanf("%f", &laps[i]);
}
```

3.2 Total set to 0 and fastest set to the first time ✓; loop through the array ✓; add to total ✓; compare and update fastest ✓; average = total ÷ 5 ✓; display both ✓, e.g. (6)

```
float total = 0, fastest = laps[0];
for (int i = 0; i < 5; i++) {
  total = total + laps[i];
  if (laps[i] < fastest) fastest = laps[i];
}
printf("Average %.2f, fastest %.2f", total / 5, fastest);
```

3.3 Loop ✓; prompt and read inside the loop ✓; condition repeats while time <= 0 ✓; correct syntax ✓, e.g. (4)

```
float t;
do {
  printf("Lap time (> 0): ");
  scanf("%f", &t);
} while (t <= 0);
```

3.4 A for loop is used when the number of repetitions is known in advance (e.g. 5 laps) ✓✓; a while loop repeats as long as a condition is true, when the number of repetitions is not known (e.g. until valid input is entered) ✓. (3)

[18]

QUESTION 4

4.1 Header with float return and float parameter ✓; correct formula ✓✓ (use 9.0 / 5 or 9 / 5.0 to avoid whole-number division); return ✓, e.g. float cToF(float c) that returns c * 9.0 / 5 + 32; (4)

4.2 Call with 25 ✓, store or use the result ✓, printf with %f / %.2f ✓, e.g. printf("%.2f F", cToF(25)); (gives 77.00 F) (3)

4.3 void return type ✓; int parameter n ✓; loop n times ✓; printf("*") ✓; new line after the loop ✓, e.g. (5)

```
void printLine(int n) {
  for (int i = 0; i < n; i++) {
    printf("*");
  }
```

```
  printf("\n");
}
```

4.4 Any TWO ✓✓: code can be reused without retyping; programs are shorter and easier to read; easier to test and fix one part at a time; team members can work on different functions. (2)

4.5 A parameter is the variable in the function's header that receives a value ✓; an argument is the actual value passed when the function is called (e.g. 25 in cToF(25)) ✓. (2)

[16]

QUESTION 5

5.1 Any SIX ✓✓✓✓✓✓: "when green flag clicked" to start; "when right / left / up / down arrow key pressed" hat blocks; "change x by 10 / −10" and "change y by 10 / −10" (or point in direction + move); "forever" loop; "if touching colour red then"; "say Battery low! for 2 seconds"; go to a start position at the beginning. (6)

5.2 Any FOUR, each identified ✓ and corrected ✓: Main → main; missing semicolon: int count = 0; i is not declared: for (int i = 0; …); missing closing brace } after count = count + 1; (end of the loop); %f should be %d for an integer. (8)

5.3 5 ✓✓ (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE